FlashDrop – Advanced Smart Delivery & Shop Schedule Management System (Improved Version)

Update my existing FlashDrop Android application built with Kotlin, Jetpack Compose, Firebase Authentication, and Cloud Firestore.

IMPORTANT: First inspect the existing project and its implementation. Do not rebuild the application or remove existing features. Improve and integrate the following functionality into the existing system.

1. Accurate Expected Delivery Time

Repair the existing expected delivery time calculation.

The default delivery estimate must be exactly 30 minutes after the actual current time.

Examples:

Current Time| Expected Delivery
9:28 PM| 9:58 PM
11:45 PM| 12:15 AM (Tomorrow)
8:20 AM| 8:50 AM

Requirements:

- Use the correct local Indian timezone (Asia/Kolkata).
- Automatically calculate 30 minutes forward.
- Correctly handle midnight and date changes.
- Never display incorrect time ranges.
- Refresh the estimate when required.
- Respect shop opening and closing schedules.
- Never promise delivery outside the shop's confirmed working hours.

2. Shopkeeper-Controlled Opening & Closing Schedule

Create a new page inside the existing Shopkeeper Dashboard named:

Shop Schedule & Auto Opening/Closing

The default schedule should be:

- Opening: 10:00 AM
- Closing: 10:00 PM

However, these are default values, NOT permanent restrictions. Every shopkeeper must be able to customize their own working hours.

Schedule Management Table

Create a proper editable table with the following settings:

Setting| Default| Action
Shop Opening Time| 10:00 AM| Edit
Shop Closing Time| 10:00 PM| Edit
Automatic Opening| ON| ON/OFF
Automatic Closing| ON| ON/OFF
Manual Shop Status| Open| Change
Current Shop Status| Automatic| View
Next Opening Time| Automatic| View
Next Closing Time| Automatic| View

Important ON/OFF Functionality

Make every ON/OFF switch fully functional.

When Automatic Opening is ON:

- Automatically open the shop at its saved opening time.
- Update customer-facing shop availability.

When Automatic Opening is OFF:

- Do not automatically open the shop.
- Require the shopkeeper to manually open it.

When Automatic Closing is ON:

- Automatically close the shop at its saved closing time.
- Disable new customer orders.

When Automatic Closing is OFF:

- Do not automatically close the shop.
- Allow the shopkeeper to manage availability manually.

Changes must be saved permanently in Firestore and remain active even after closing the application.

3. Custom Weekly Shop Schedule

Add an additional option for shopkeepers to manage different timings for different days.

For example:

Day| Opening| Closing| Status
Monday| 10:00 AM| 10:00 PM| ON

Tuesday| 10:00 AM| 10:00 PM| ON
Wednesday| 10:00 AM| 9:00 PM| ON
Thursday| 10:00 AM| 10:00 PM| ON
Friday| 10:00 AM| 10:00 PM| ON
Saturday| 10:00 AM| 11:00 PM| ON
Sunday| 10:00 AM| 10:00 PM| OFF

Allow shopkeepers to edit each day's opening and closing times independently.

Include:

- Save Weekly Schedule button.
- Enable/Disable individual days.
- Apply the saved schedule automatically every week.
- Support different opening and closing times.

4. Intelligent Late-Night Order Management

If an order arrives close to closing time, ask the shopkeeper whether it can be delivered today.

Example:

Order placed: 9:30 PM.

Show a notification in the Shopkeeper Dashboard:

Can you deliver this order today?

Provide two options:

1. DELIVER TODAY
2. DELIVER TOMORROW

If DELIVER TODAY is selected:

- Ask the shopkeeper to confirm the delivery estimate.
- Ensure the confirmed delivery time is within the shop's operating schedule.
- Show the confirmed estimate to the customer.

If DELIVER TOMORROW is selected:

- Automatically schedule delivery for the next working day.
- Start the delivery calculation from the next opening time.
- Calculate 30 minutes forward from that opening time.

Example:

Order placed: 9:30 PM
Shop closes: 10:00 PM
Next opening: 10:00 AM Tomorrow
Expected delivery: 10:30 AM Tomorrow.

If the next day is a scheduled holiday or the shop is disabled for that day, automatically select the next available working day.

5. Customer-Side Shop Availability

Update the existing customer interface.

Display:

- GREEN OPEN label when the shop is available.
- CLOSED label when the shop is unavailable.
- Next opening time when closed.
- Expected delivery date and time.
- Clear notification when delivery is scheduled for tomorrow.

When a shop is closed:

- Disable Add to Cart/Order Now actions for that shop.
- Prevent checkout for unavailable shops.
- Display a clear explanation to customers.

Do not remove existing product listings, cart, checkout, or order tracking.

6. Firebase Database Integration

Permanently save the following information for each shop:

- Opening time.
- Closing time.
- Weekly schedule.
- Automatic opening ON/OFF.
- Automatic closing ON/OFF.
- Current availability.
- Manual open/close status.
- Temporary closure.
- Next opening time.
- Next closing time.
- Last schedule update.
- Shopkeeper confirmation for late-night orders.

Each shopkeeper must only be able to modify their own shop's settings.

Use Firebase Security Rules and trusted backend validation.

Use Firebase Cloud Functions and Cloud Scheduler for automatic opening and closing, so the functionality continues even when the shopkeeper's application is closed.

Do not depend exclusively on the mobile device clock.

7. Admin Panel Integration

Both existing Admin panels must be able to view shop operating schedules and current availability.

Allow authorized administrators to:

- View every shop's opening and closing schedule.
- View automatic opening/closing settings.
- View current shop availability.
- Temporarily disable a shop when necessary.
- View schedule changes.

Do not allow administrators to accidentally overwrite shopkeeper schedules without an explicit action.

8. Design and Responsiveness

Maintain the existing FlashDrop branding:

- Green and orange colour theme.
- Existing logo.
- Existing buttons and navigation style.

Make the new schedule page responsive on:

- Small phones.
- Large phones.
- Tablets.

Use properly spaced tables, readable text, accessible ON/OFF switches, and appropriate padding.

Do not modify unrelated application screens or redesign the entire application.

9. Testing Requirements

Test the following scenarios:

1. Shop automatically opens at 10:00 AM.
2. Shop automatically closes at 10:00 PM.
3. Shopkeeper changes opening and closing times.

4. Automatic Opening is switched OFF.
5. Automatic Closing is switched OFF.
6. Shopkeeper manually closes the shop.
7. Shopkeeper changes the weekly schedule.
8. Customer tries ordering when the shop is closed.
9. Customer places an order at 9:30 PM.
10. Shopkeeper chooses Deliver Tomorrow.
11. Next day is a weekly holiday.
12. Expected delivery crosses midnight.
13. Application is closed during automatic opening/closing.
14. Schedule remains saved after logout and login.
15. Multiple shops operate independently.

FINAL INSTRUCTION

Implement the complete functionality, not just the user interface.

Most importantly:

- Keep the shopkeeper's opening/closing time table.
- Make all ON/OFF switches genuinely functional.
- Allow every shopkeeper to customize their schedule.
- Automatically apply saved schedules.
- Keep all settings permanently saved in Firebase.
- Make delivery calculations respect actual shop availability.
- Integrate the system into the existing Customer, Shopkeeper, and Admin panels.

Do not replace existing working functionality with dummy data or temporary local-only storage. Do not break existing Firebase authentication, database structure, order management, or other FlashDrop features.

First inspect, then implement, then test the complete system.